Md Kamruzzaman

AI & Backend Engineer | MLOps | FastAPI | Node.js

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, Machine Learning, 2025) with hands-on experience building and deploying backend APIs and ML serving pipelines using Python, FastAPI, and Node.js. Developed production-style personal projects featuring REST API design, JWT authentication, database optimization, Docker, and Retrieval-Augmented Generation (RAG). Seeking a remote Backend or MLOps role to contribute to scalable systems.

EXPERIENCE

Backend & AI Developer (Independent Projects) Mar 2026 – Present

NoteVault | *Secure Note Management System* | Live Demo · GitHub Oct 2026

- Developed a secure full-stack application within a 48-hour deadline, successfully adapting Python/FastAPI architectural concepts to a Node.js, Express, and MongoDB stack.
- Implemented JWT authentication, bcrypt password hashing, and Role-Based Access Control (RBAC) for distinct User and Admin roles.
- Designed compound MongoDB indexes (`{owner: 1, createdAt: -1}`) to optimize pagination and built aggregation pipelines using `$unwind`, `$group`, and `$lookup`.

Technologies: ***Node.js, Express, MongoDB, React, JWT***

Expense Tracker API | *Backend Engineering & CI/CD* | Live API · GitHub Mar 2026 – Oct 2026

- Engineered a secure REST API with 8 endpoints using FastAPI, enforcing user-scoped data isolation and strict data validation.
- Achieved **91% test coverage** using pytest and integrated GitHub Actions for continuous integration (CI).
- Containerized the application with Docker to ensure consistent development and deployment environments.

Technologies: ***Python, FastAPI, PostgreSQL, SQLAlchemy, Docker, GitHub Actions, pytest***

Docu-RAG API | *Generative AI & LLM Integration* | GitHub Oct 2026 – Present

- Developed a Retrieval-Augmented Generation (RAG) API for context-aware Q&A over PDF documents, achieving a **2–4 second end-to-end latency**.
- Engineered a document ingestion pipeline using LangChain's `RecursiveCharacterTextSplitter` to generate and store vector embeddings.
- Integrated the Google Gemini API with ChromaDB to ground answers in document context and reduce LLM hallucinations.

Technologies: ***Python, FastAPI, LangChain, ChromaDB, Google Gemini API, Docker***

ML Expense Platform | *MLOps & Model Serving* | Live API · GitHub Sep 2026 – Oct 2026

- Architected an end-to-end ML serving pipeline with FastAPI, scikit-learn, and MLflow to classify free-text expense data.
- Achieved **~5ms inference latency** by loading the model into memory at API startup; handled 500+ simulated requests per second.
- Containerized the service with Docker and added Prometheus metrics and Grafana dashboards for real-time monitoring.

Technologies: ***Python, scikit-learn, MLflow, FastAPI, Docker, Kubernetes, Prometheus, Grafana***

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering (Machine Learning) | *Lovely Professional University* 2025

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript (Node.js), Bash
- **Backend & APIs:** FastAPI, REST APIs, SQLAlchemy, Pydantic, Express.js, JWT, RBAC
- **Databases:** PostgreSQL, MongoDB, Mongoose, ChromaDB (Vector Database)
- **AI & MLOps:** LLM Applications, RAG Pipelines, LangChain, Google Gemini API, MLflow, scikit-learn
- **Cloud & DevOps:** Docker, Kubernetes, GitHub Actions (CI/CD), Prometheus, Grafana, Render, Vercel
- **Core Concepts:** System Design, Data Structures, Algorithms

CERTIFICATIONS & OPEN SOURCE

- **Certifications:** Natural Language Processing Specialization (DeepLearning.AI), Meta Back-End Developer (Coursera), Python Data Structures (University of Michigan)
- **Open Source:** Maintain 6 public GitHub repositories with 80+ commits, focused on Python backend architecture and GenAI applications